

A Full Affirmative Answer to Kerr–Li Question 9.4

Simple AF Automorphisms with Arbitrary Prescribed CA Entropy

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Abstract

Kerr and Li asked whether a UHF algebra, or any other simple AF algebra, can admit a $*$ -automorphism with finite nonzero contractive approximation (CA) entropy. We give an affirmative answer to the “any other simple AF algebra” part, in a stronger form: every value in $[0, \infty]$ occurs as the CA entropy of an automorphism of a simple unital separable AF algebra.

Given a minimal homeomorphism T of a Cantor set X , consider the inductive system

$$A_n = M_{2^{n-1}}(C(X)), \quad \varphi_n(a)(x) = \text{diag}(a(x), a(Tx)).$$

The maps $\beta_n(a) = a \circ T$ are compatible with the connecting maps and induce an automorphism β of the limit. Minimality of T makes the limit simple, while zero-dimensionality of X makes it AF. A locality property of contractive ranks and Kerr–Li’s injective-tensor-product subadditivity show

$$h_c(\beta) = h_{\text{top}}(T).$$

Minimal Cantor systems exist with every prescribed entropy in $[0, \infty]$.

We also audit the supplied product-type argument. Its mathematical core is correct, with two minor corrections: the cited source result is Proposition 2.1 rather than Proposition 3.1, and for $p > 2$ the displayed Pauli matrices in an M_2 corner of M_p are norm-one self-adjoint elements, not unitaries of M_p .

1 The question and the answer

Kerr and Li proved that the bilateral tensor shift on $M_p^{\otimes \mathbb{Z}}$ has infinite CA entropy and then asked:

Is there a $*$ -automorphism of $M_p^{\otimes \mathbb{Z}}$ (or any other simple AF algebra) with finite nonzero CA entropy?

The following theorem answers the second alternative affirmatively.

Theorem 1.1 (Main theorem). *Let X be a Cantor set and let $T: X \rightarrow X$ be a minimal homeomorphism. There are a simple unital separable AF algebra A and an automorphism $\beta \in \text{Aut}(A)$ such that*

$$h_c(\beta) = h_{\text{top}}(T).$$

Corollary 1.2. *For every $r \in [0, \infty]$ there are a simple unital separable AF algebra A_r and an automorphism $\beta_r \in \text{Aut}(A_r)$ satisfying*

$$h_c(\beta_r) = r.$$

In particular, simple AF algebras admit automorphisms with finite nonzero CA entropy.

The proof is elementary once three permanence observations are isolated.

Remark 1.3 (Scope). The construction settles the alternative “or any other simple AF algebra. It does not by itself decide whether a prescribed UHF algebra $M_p^{\otimes \mathbb{Z}}$ admits an automorphism with finite nonzero CA entropy.

2 Locality and invariant inductive limits

We use Kerr–Li’s notation $\text{rc}_X(\Omega, \delta)$ for the contractive rank of a finite set Ω in a Banach space X .

Lemma 2.1 (Locality of contractive rank). *Let Y be a closed subspace of a Banach space X . If $\Omega \subseteq Y$ is finite and $\delta > 0$, then*

$$\text{rc}_X(\Omega, \delta) = \text{rc}_Y(\Omega, \delta).$$

Consequently, if α is an isometric automorphism of X , Y is α -invariant, and $\Omega \subseteq Y$, then

$$\text{h}_c(\alpha, \Omega) = \text{h}_c(\alpha|_Y, \Omega).$$

Proof. Choose a CA embedding $\iota: X \rightarrow Z$. Its restriction to Y is a CA embedding. Restricting the first map in any contractive approximation for X gives $\text{rc}_Y(\Omega, \delta) \leq \text{rc}_X(\Omega, \delta)$.

Conversely, suppose

$$Y \xrightarrow{\phi} \ell_\infty^d \xrightarrow{\psi} Z$$

is a contractive approximation to $\iota|_Y$ on Ω . The space ℓ_∞^d is injective, so ϕ extends to a contraction $\tilde{\phi}: X \rightarrow \ell_\infty^d$. The pair $(\tilde{\phi}, \psi)$ gives the same approximation on Ω , proving the reverse inequality. Applying this equality to orbit sets gives the entropy assertion. $\square$

Proposition 2.2 (Increasing invariant unions). *Let $X_1 \subseteq X_2 \subseteq \dots$ be closed invariant subspaces of a Banach space X with dense union, and let α be an isometric automorphism of X preserving every X_n . Then*

$$\text{h}_c(\alpha) = \sup_n \text{h}_c(\alpha|_{X_n}).$$

Proof. The inequality “ $\geq$ ” is monotonicity. For the converse, choose an increasing sequence of finite subsets of $\bigcup_n X_n$ whose union is dense. Each such finite set is contained in one X_n , and its local entropy in X equals its local entropy in X_n by Lemma 2.1. Kerr–Li’s Kolmogorov–Sinai-type exhaustion property, Proposition 2.1(ii) of [4], now gives the result. $\square$

3 Finite matrix amplification of a Cantor system

For a homeomorphism $T: X \rightarrow X$, write $\alpha_T(f) = f \circ T$ on $C(X)$ and let

$$\alpha_T^{(d)} = \alpha_T \otimes \text{id}_{M_d}$$

under the identification $C(X, M_d) = C(X) \otimes M_d$.

Proposition 3.1. *For every compact Hausdorff space X , every homeomorphism T of X , and every $d \geq 1$,*

$$\text{h}_c(\alpha_T^{(d)}) = \text{h}_{\text{top}}(T).$$

Proof. The invariant scalar subalgebra $C(X) \otimes 1_d$ gives

$$h_c(\alpha_T^{(d)}) \geq h_c(\alpha_T) = h_{\text{top}}(T),$$

where the equality is Kerr–Li’s Proposition 3.1.

As Banach spaces,

$$C(X, M_d) = C(X) \check{\otimes} M_d,$$

since one tensor factor is commutative (and M_d is finite-dimensional). Kerr–Li’s injective-tensor-product subadditivity, Proposition 7.1, yields

$$h_c(\alpha_T^{(d)}) \leq h_c(\alpha_T) + h_c(\text{id}_{M_d}) = h_{\text{top}}(T),$$

because every automorphism of a finite-dimensional Banach space has zero CA entropy. $\square$

4 Construction of the simple AF algebra

Fix a minimal homeomorphism $T: X \rightarrow X$ of a Cantor set. For $n \geq 1$ put

$$A_n = M_{2^{n-1}}(C(X)) = C(X, M_{2^{n-1}})$$

and define a unital injective $*$ -homomorphism

$$\varphi_n: A_n \longrightarrow A_{n+1}, \quad \varphi_n(a)(x) = \begin{pmatrix} a(x) & 0 \\ 0 & a(Tx) \end{pmatrix}.$$

Set

$$A = \varinjlim (A_n, \varphi_n).$$

4.1 The limit is AF

Since X is zero-dimensional, locally constant functions are dense in $C(X)$. Thus $C(X)$ is AF, each A_n is AF, and an inductive limit of AF algebras is AF. The maps are unital and the system is countable, so A is unital and separable.

4.2 The limit is simple

For $m > n$, iteration of the connecting maps gives, up to a permutation of matrix blocks,

$$\varphi_{n,m}(a)(x) \cong \bigoplus_{j=0}^{m-n} a(T^j x)^{\oplus \binom{m-n}{j}}. \quad (1)$$

Indeed, each further application replaces every block $b(x)$ by the two blocks $b(x)$ and $b(Tx)$, producing Pascal’s rule.

Let $0 \neq a \in (A_n)_+$ and put

$$U = \{x \in X : a(x) \neq 0\}.$$

This is a nonempty open set. Minimality implies that the forward orbit of every point meets U , and compactness gives an $L \geq 0$ such that

$$X = \bigcup_{j=0}^L T^{-j}(U).$$

Taking $m \geq n + L$, formula (1) shows that $\varphi_{n,m}(a)(x) \neq 0$ for every $x \in X$. A positive element of $C(X, M_k)$ is full exactly when it is nonzero at every point. Hence every nonzero positive element at any stage becomes full at a later stage. The standard simplicity criterion for injective inductive limits therefore shows that A is simple.

4.3 The compatible automorphism

Define

$$\beta_n(a)(x) = a(Tx), \quad a \in A_n.$$

Then

$$\varphi_n \circ \beta_n = \beta_{n+1} \circ \varphi_n,$$

since both sides send a to $\text{diag}(a(Tx), a(T^2x))$. The β_n therefore induce an automorphism $\beta \in \text{Aut}(A)$.

By Proposition 3.1,

$$h_c(\beta_n) = h_{\text{top}}(T) \quad \text{for every } n.$$

The images of the A_n form an increasing family of closed β -invariant subspaces with dense union. Proposition 2.2 now gives

$$h_c(\beta) = \sup_n h_c(\beta_n) = h_{\text{top}}(T).$$

This proves Theorem 1.1.

Proof of Corollary 1.2. Boyle and Handelmann constructed minimal homeomorphisms of Cantor sets with every prescribed topological entropy in $[0, \infty]$; in fact these may all be chosen strongly orbit equivalent to the dyadic adding machine [1]. Apply Theorem 1.1. $\square$

5 Audit of the supplied product-type partial result

The supplied theorem for factor-permuting product-type automorphisms is correct. Here are the points that need correction or clarification.

- (1) The monotonicity and exhaustion result used in the zero-entropy case is Kerr–Li Proposition 2.1, not Proposition 3.1.
- (2) If $p > 2$, the two Pauli matrices placed in the upper-left M_2 corner of M_p are not unitaries of M_p : they have an additional zero block. They are, however, norm-one self-adjoint elements, and for $x = au + bv$ with $a^2 + b^2 = 1$ the spectrum contains 1. This is all that the argument needs.
- (3) In the infinite-orbit case, the product-state step can be stated without invoking an infinite product state. The finitely many distinct tensor factors generated by the terms form a full matrix algebra. Take the tensor product of states attaining value 1 on the chosen summands, and extend that state from the finite-dimensional subalgebra to the whole C^* -algebra.

With these edits, the proof gives the claimed dichotomy 0 or ∞ for that product-type class. It is fully compatible with the construction above: our automorphism does not permute a fixed tensor decomposition into matrix factors; it is assembled from diagonal orbit embeddings of matrix-valued Cantor function algebras.

6 Literature status

Kerr–Li’s paper itself gives arbitrary CA entropy values on the simple Bunce–Deddens algebra, but that algebra is AT rather than AF. The most direct later AF result located in the literature search is Kerr’s 2007 paper [3], which constructs a simple unital AF algebra for which every non-approximately-inner automorphism has infinite CA entropy. That result does not provide a finite positive example.

Exact-phrase searches, citation-chain searches around [4] and [3], and a check of the recent revisit of noncommutative approximation entropy in [2] did not locate a published resolution of Question 9.4. Thus no priority claim is made for the argument above, but it appears to supply the missing affirmative construction.

References

- [1] M. Boyle and D. Handelman, *Entropy versus orbit equivalence for minimal homeomorphisms*, Pacific J. Math. **164** (1994), 1–13.
- [2] B. Jacelon and R. Neagu, *Brown–Voiculescu entropy revisited*, arXiv:2506.18352, version 2 (2026).
- [3] D. Kerr, *Generically infinite entropy in a simple AF algebra*, Bull. Lond. Math. Soc. **39** (2007), 265–271.
- [4] D. Kerr and H. Li, *Dynamical entropy in Banach spaces*, Invent. Math. **162** (2005), 649–686; arXiv:math/0407386.